

Mathematics A

Unit 1 • Chapter OL-T43 Classified

Cross-session • chapter-organised candidate

15 classified items • 62 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Unit 1 • Chapter OL-T43 • 15 items • 62 marks

Chapter	Items	Marks	Starts
01. OL-T43 Area & Perimeter	15	62	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Area & Perimeter

Unit 1 • 62 marks • 15 item(s)

June 2025 | 4MA1-1H Q9 | 5 marks

Combine polygon area with Pythagoras or trigonometry

June 2024 | 4MA1-2H Q7 | 4 marks

Recover a missing trapezium or parallelogram dimension

November 2023 | 4MA1-2H Q10 | 5 marks

Find composite or shaded polygonal area

June 2022 | 4MA1-2H Q5 | 4 marks

Recover a missing trapezium or parallelogram dimension

January 2022 | 4MA1-1H Q9 | 4 marks

Combine polygon area with Pythagoras or trigonometry

January 2020 | 4MA1-1H Q3 | 4 marks

Recover a missing dimension from area

January 2020 | 4MA1-1H Q14 | 2 marks

Find area of a parallelogram

January 2021 | 4MA1-2H Q18 | 3 marks

Find perimeter of a simple triangle or rectangle

January 2021 | 4MA1-2H Q20 | 4 marks

Find composite or shaded polygonal area

June 2021 | 4MA1-1H Q3 | 4 marks

Recover a missing dimension from area

November 2021 | 4MA1-1H Q4 | 5 marks

Use an area result in a real-world quantity or cost

November 2021 | 4MA1-2H Q10 | 4 marks

Find perimeter of a composite polygonal shape

May-Nov 2020 | 1H Q11 | 6 marks

Recover a missing dimension from area

May-Nov 2020 | 2H Q8 | 4 marks

Recover a missing dimension from area

January 2023 | 2H Q4 | 4 marks

Find perimeter of a simple triangle or rectangle

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

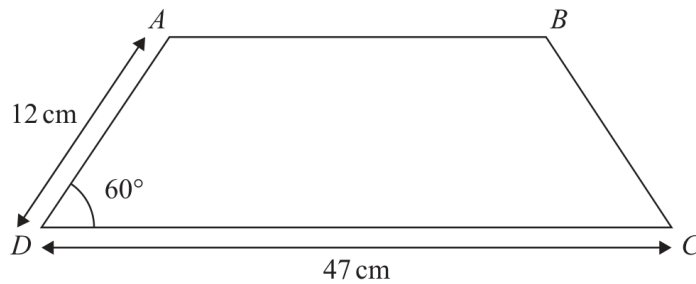
Working Unit 06

4MA1-1H Q9

ORIGINAL QUESTION

5 marks

9

Diagram **NOT**
accurately drawn $ABCD$ is a trapezium with one line of symmetry.

$$\text{angle } ADC = 60^\circ \quad AD = 12 \text{ cm} \quad DC = 47 \text{ cm}$$

Work out the area of the trapezium.

Give your answer correct to 3 significant figures.

Show your working clearly.

..... cm^2

(Total for Question 9 is 5 marks)

Worked Solution 06

Area of a symmetric trapezium

UNIT 1
5 marks**Height and horizontal projection**

$$h = 12 \sin 60^\circ = 6\sqrt{3}, \quad 12 \cos 60^\circ = 6.$$

Upper base

$$AB = 47 - 6 - 6 = 35 \text{ cm.}$$

Area

$$A = \frac{1}{2}(47 + 35)(6\sqrt{3}) = 246\sqrt{3} = 426.084 \dots$$

Final answer

$$426 \text{ cm}^2 \text{ (3 s.f.)}$$

Dependency check The line of symmetry makes the two horizontal projections equal.

Question 21

4MA1-2H Q7

ORIGINAL QUESTION

4 marks

- 7 The diagram shows a 6-sided shape $ABCDEF$

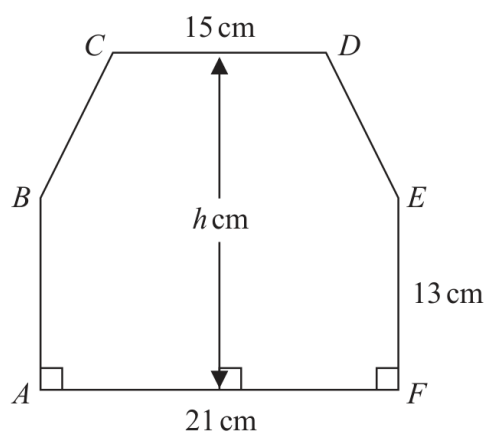


Diagram **NOT**
accurately drawn

$$AF = 21 \text{ cm} \quad CD = 15 \text{ cm} \quad AB = FE = 13 \text{ cm}$$

The perpendicular height of the shape is h cm

CD is parallel to AF

The area of the shape is 390 cm^2

Work out the value of h

$h = \dots\dots\dots$

Worked Solution 21

Composite area

MODULAR UNIT 1

4 marks

Area equation

$$21(13) + \frac{1}{2}(21 + 15)(h - 13) = 390.$$

Solve

$$273 + 18(h - 13) = 390 \Rightarrow h = 19.5.$$

Final answer

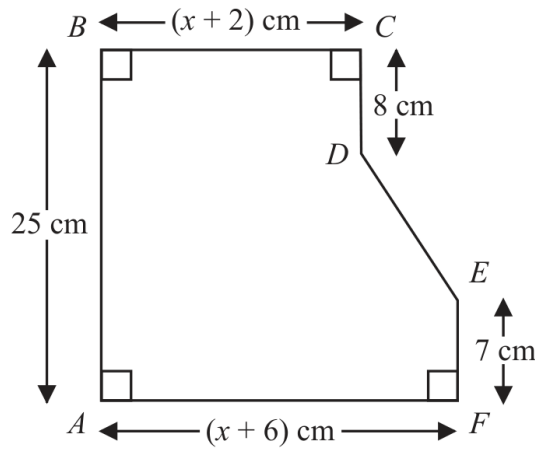
19.5 cm

Question 22

4MA1-2H Q10 whole

ORIGINAL QUESTION UNIT 1

5 marks

10 The diagram shows a hexagon $ABCDEF$ Diagram **NOT**
accurately drawn

$$AB = 25 \text{ cm} \quad BC = (x + 2) \text{ cm} \quad CD = 8 \text{ cm} \quad EF = 7 \text{ cm} \quad AF = (x + 6) \text{ cm}$$

The area of hexagon $ABCDEF$ is 258 cm^2 Work out the value of x $x = \dots\dots\dots$

Worked solution 22

Area of a stepped hexagon

MODULAR UNIT 1

5 marks

whole – Area of a stepped hexagon

Read the missing vertical lengths

DE drops $25 - 8 - 7 = 10$ cm; the horizontal width of the sloping strip is $(x + 6) - (x + 2) = 4$ cm.

Split into a rectangle and trapezium

$$A = 25(x + 2) + \frac{1}{2}(17 + 7)(4) = 25(x + 2) + 48.$$

Use the given area

$$25(x + 2) + 48 = 258 \Rightarrow 25x + 98 = 258.$$

Solve

$$25x = 160 \Rightarrow x = 6.4.$$

Final answer

$$x = 6.4$$

Original question 18

4MA1-2H Q5 Q5

ORIGINAL QUESTION**4 marks**

- 5 The diagram shows an 8-sided shape $ABCDEFGH$.

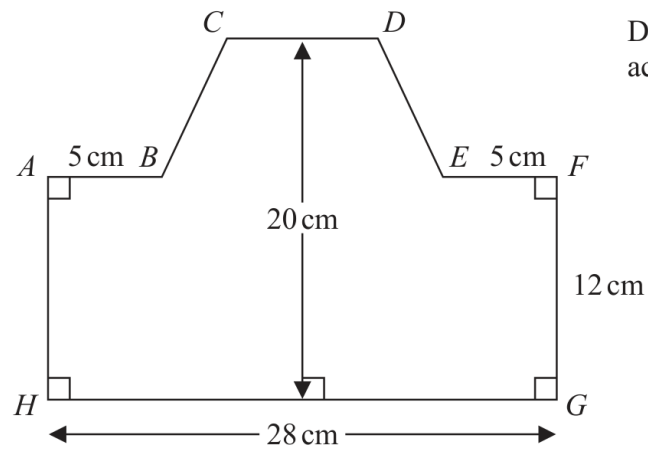


Diagram **NOT**
accurately drawn

$$HG = 28 \text{ cm} \quad FG = 12 \text{ cm} \quad AB = EF = 5 \text{ cm}$$

The height of the shape is 20 cm

CD is parallel to HG

The area of shape $ABCDEFGH$ is 434 cm^2

Find the length of CD .

..... cm

Worked solution 18

Chapter: Area & Perimeter

MODULAR UNIT 1

4 marks

Q5 – Chapter: Area & Perimeter

Family: Use parallelogram and trapezium area • Variant: Recover a missing trapezium or parallelogram dimension

Method

The lower rectangle has area $28 \times 12 = 336 \text{ cm}^2$.

Why: Its width and height are given by HG and FG.

Method

The top section has height $20 - 12 = 8 \text{ cm}$ and lower parallel side $28 - 5 - 5 = 18 \text{ cm}$.

Why: Remove the two 5 cm horizontal offsets.

Method

If $CD = x$, then $336 + \frac{1}{2}(18 + x) \times 8 = 434$.

Why: Add the rectangle area to the trapezium area.

Method

$4(18 + x) = 98$, so $x = 6.5$ and $CD = 6.5 \text{ cm}$.

Why: Solve the resulting linear equation.

Final answer

CD: $CD = 6.5 \text{ cm}$

Original question 07

4MA1-1H Q9

ORIGINAL QUESTION**4 marks**

- 9 The diagram shows isosceles triangle ABC

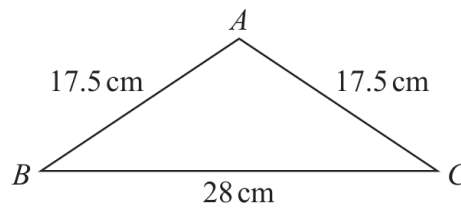


Diagram **NOT**
accurately drawn

$$AB = AC = 17.5 \text{ cm}$$

$$BC = 28 \text{ cm}$$

Calculate the area of triangle ABC

..... cm^2

Worked solution 07

Chapter: Area & Perimeter

MODULAR UNIT 1

4 marks

Q9 – Chapter: Area & Perimeter

Family: Use parallelogram and trapezium area • Variant: Combine polygon area with Pythagoras or trigonometry

Method

The perpendicular height is $\sqrt{(17.5^2 - 14^2)} = \sqrt{110.25} = 10.5$ cm.

Why: Bisect the 28 cm base of the isosceles triangle and use Pythagoras.

Method

Area = $\frac{1}{2} \times 28 \times 10.5$.

Why: Use base times perpendicular height divided by 2.

Method

Area = 147 cm^2 .

Why: Evaluate the product.

Method

Therefore the area is 147 cm^2 .

Why: State the requested area.

Final answer

area: 147 cm^2

3 The diagram shows a shape.

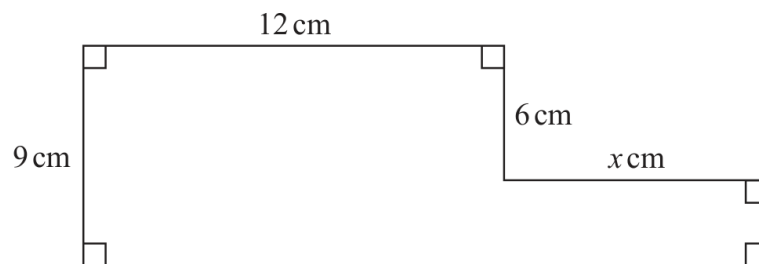


Diagram **NOT**
accurately drawn

The shape has area 129 cm^2

Work out the value of x .

$x =$

Worked solution

January 2020 | Question 3 | 4 marks

Family: Use rectangle and triangle area • Variant: Recover a missing dimension from area

Method / working

1. The left rectangle has area $12 \times 9 = 108$.
2. The right strip has height $9 - 6 = 3$ and area $3x$.
3. $108 + 3x = 129$, so $3x = 21$.
4. Therefore $x = 7$ cm.

Final answer: $x = 7$ cm

14 The diagram shows parallelogram $EFGH$.

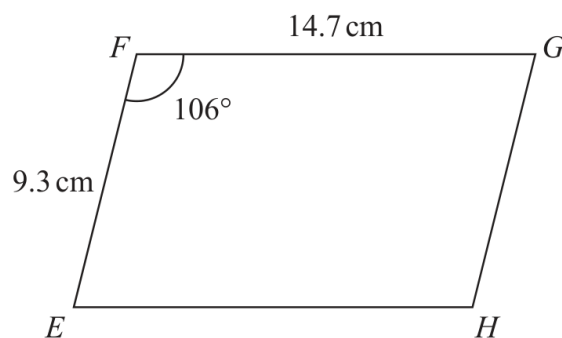


Diagram **NOT**
accurately drawn

$$EF = 9.3\text{ cm}$$

$$FG = 14.7\text{ cm}$$

$$\text{Angle } EFG = 106^\circ$$

- (a) Work out the area of the parallelogram.
Give your answer correct to 3 significant figures.

..... cm^2
(2)

Worked solution

January 2020 | Question 14 | 2 marks

Family: Use parallelogram and trapezium area • Variant: Find area of a parallelogram

Method / working

1. $\text{Area} = ab \sin \theta = 9.3 \times 14.7 \times \sin 106^\circ$.

2. $\text{Area} = 131.414 \dots \sim 131 \text{ cm}^2$ (3 s.f.).

Final answer: 131 cm^2

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$

B has coordinates $(1, 10)$

C has coordinates $(19, a)$

D has coordinates (b, c)

(a) Work out the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

..... cm

(3)

Worked solution

January 2021 | Question 18 | 3 marks

Family: Find perimeter of polygonal shapes • Variant: Find perimeter of a simple triangle or rectangle

Method / working

1. $AB = \sqrt{5^2 + 12^2} = 13 \text{ cm.}$
2. $BC = \sqrt{18^2 + 7.5^2} = 19.5 \text{ cm.}$
3. $\text{Perimeter} = 2(13 + 19.5) = 65 \text{ cm.}$

Final answer: 65 cm

20

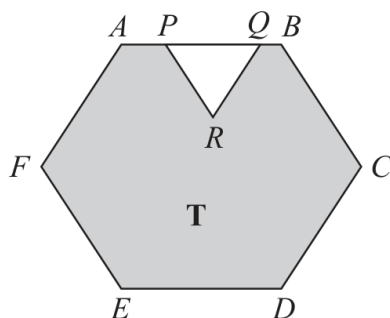


Diagram **NOT**
accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3} \text{ cm}^2$

work out the length of PQ .

..... cm

Worked solution

January 2021 | Question 20 | 4 marks

Family: Use rectangle and triangle area • Variant: Find composite or shaded polygonal area

Method / working

1. Let $PQ=x$, so $AB=1.5x$.
2. Write the equilateral triangle and regular-hexagon areas in terms of x .
3. Set shaded area equal to $72\sqrt{3}$ and solve the resulting quadratic in x^2 .
4. The positive length is $PQ=4.8$ cm.

Final answer: 4.8 cm

3

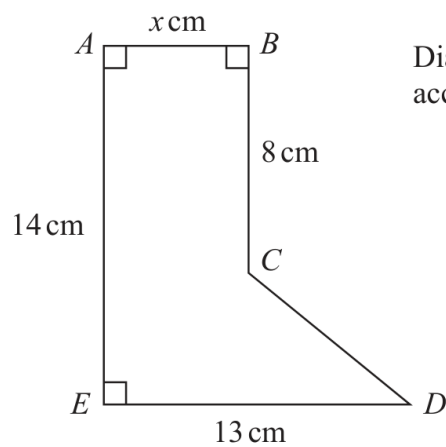


Diagram **NOT**
accurately drawn

The diagram shows the shape $ABCDE$.

The area of the shape is 91.8 cm^2

Work out the value of x .

$x = \dots\dots\dots$

Worked solution

June 2021 | Question 3 | 4 marks

Family: Use rectangle and triangle area • Variant: Recover a missing dimension from area

Method / working

1. Write the area equation for the composite shape: $11x + 39 = 91.8$.
2. Subtract 39: $11x = 52.8$.
3. Divide by 11 to get $x = 4.8$.
4. Check the positive length and give cm.

Final answer: 4.8 cm

- 4 The diagram shows the front of a wooden door with a semicircular glass window.

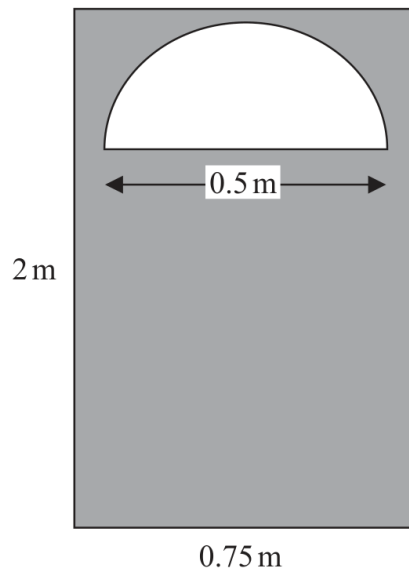


Diagram **NOT**
accurately drawn

Julie wants to apply 2 coats of wood varnish to the front of the door, shown shaded in the diagram.

250 millilitres of wood varnish covers 4 m^2 of the wood.

Work out how many millilitres of wood varnish Julie will need.
Give your answer correct to the nearest millilitre.

..... millilitres

Worked solution

November 2021 | Question 4 | 5 marks

Family: Use rectangle and triangle area • Variant: Use an area result in a real-world quantity or cost

Method / working

1. Area of the rectangular door:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $0.75 \times 2 = 1.5 \text{ m}^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. The semicircular window has diameter (0.5 m) , so

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $r = 0.25 \text{ m}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. Area of the semicircular window:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $\frac{1}{2} \pi (0.25)^2 = 0.098174 \dots \text{ m}^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. Shaded area for one coat:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $1.5 - 0.098174 \dots = 1.401825 \dots \text{ m}^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. For 2 coats:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $2 \times 1.401825 \dots = 2.80365 \dots \text{ m}^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. Since 250 ml covers (4 m^2) , varnish needed is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. $\frac{2.80365 \dots}{4} \times 250 = 175.22 \dots$

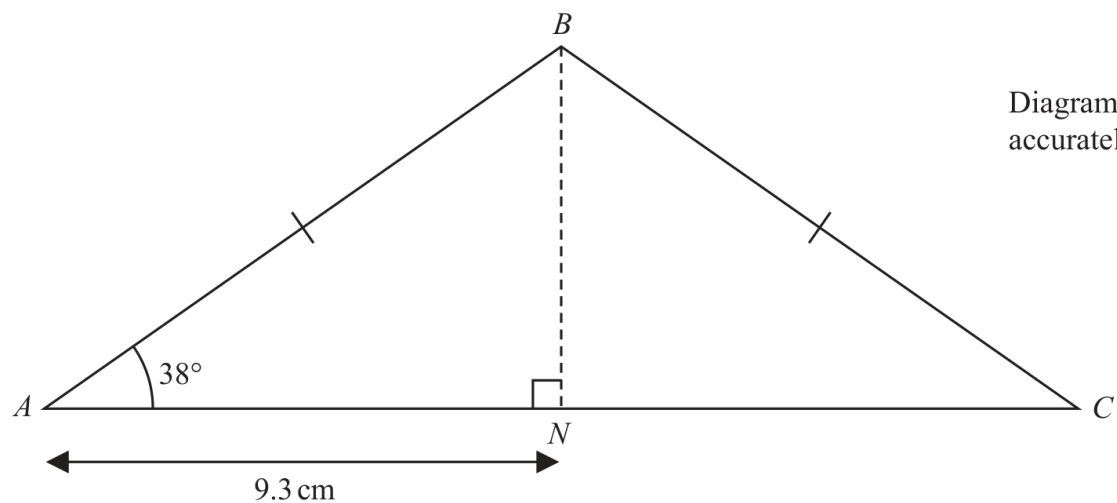
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. Answer: (175) millilitres.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: 175 millilitres.

10 ABC is an isosceles triangle with $BA = BC$.



N is the point on AC such that $AN = 9.3\text{ cm}$ and BN is perpendicular to AC .

Work out the perimeter of triangle ABC .

Give your answer correct to 3 significant figures.

..... cm

Worked solution

November 2021 | Question 10 | 4 marks

Family: Find perimeter of polygonal shapes • Variant: Find perimeter of a composite polygonal shape

Method / working

1. In right-angled triangle (ABN) ,

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $\cos 38^\circ = \frac{9.3}{AB}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $AB = \frac{9.3}{\cos 38^\circ} = 11.802 \dots$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. Since (ABC) is isosceles,

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. $BC = AB$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. Also,

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $AC = 2 \times 9.3 = 18.6$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. Perimeter:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. $11.802 \dots + 11.802 \dots + 18.6 = 42.204 \dots$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. Answer: (42.2 cm) .

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: 42.2 cm .

- 11 The diagram shows trapezium $ABCD$ in which BC and AD are parallel.

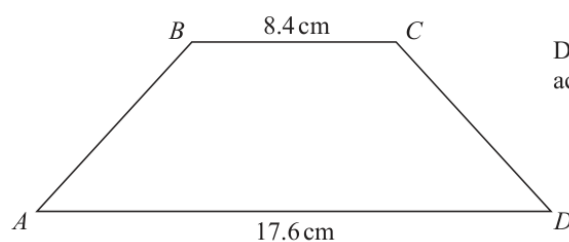


Diagram **NOT**
accurately drawn

The trapezium has exactly one line of symmetry.

$$BC = 8.4 \text{ cm}$$

$$AD = 17.6 \text{ cm}$$

The trapezium has area 179.4 cm^2

Work out the size of angle ABC .

Give your answer correct to 1 decimal place.

(Total for Question 11 is 6 marks)

Answer reference | May-Nov 2020 | Question 11

Family: Use rectangle and triangle area • Variant: Recover a missing dimension from area

11	$(AX \Rightarrow) (17.6 - 8.4) \div 2 (= 4.6)$		6	M1 where X is the foot of the perpendicular from B to AD
	$0.5 \times (8.4 + 17.6) \times h = 179.4$ or $0.5 \times '4.6' \times h + 0.5 \times '4.6' \times h + 8.4 \times h = 179.4$ or $13 \times h = 179.4$			M1
	$(h \Rightarrow) 179.4 \div '13' (=13.8)$ or $(h \Rightarrow) 358.8 \div '26' (=13.8)$ oe			M1
	$\tan ABX = \frac{'4.6'}{'13.8'}$ or $\tan BAX = \frac{'13.8'}{'4.6'}$			M1 ft their h dep on second M1 $(AB \Rightarrow) \sqrt{'4.6'^2 + '13.8'^2} = \sqrt{211.6}$ $= (14.546\dots)$ and one from $\sin ABX = \frac{'4.6'}{\sqrt{211.6}}$ or $\sin BAX = \frac{'13.8'}{\sqrt{211.6}}$ or $\cos ABX = \frac{'13.8'}{\sqrt{211.6}}$ or $\cos BAX = \frac{'4.6'}{\sqrt{211.6}}$ or $\sin ABX = \frac{'4.6' \times \sin 90}{\sqrt{211.6}}$ or $\cos ABX = \frac{\sqrt{211.6} \times '13.8'^2 - '4.6'^2}{2 \times \sqrt{211.6} \times '13.8'}$
	$(ABX \Rightarrow) \tan^{-1} \left(\frac{'4.6'}{'13.8'} \right) (= 18.4)$ or $(BAX \Rightarrow) \tan^{-1} \left(\frac{'13.8'}{'4.6'} \right) (= 71.6)$			M1
		108.4		A1 awrt 108.4
				Total 6 marks

Worked solution

May-Nov 2020 | Question 11 | 6 marks

Family: Use rectangle and triangle area • Variant: Recover a missing dimension from area

Solution

The average of the parallel sides is

$$(8.4+17.6)/(2)=13$$

So the perpendicular height is

$$179.4 / 13=13.8$$

The horizontal offset on each side is

$$(17.6-8.4)/(2)=4.6$$

The acute angle between AB and the horizontal is

$$\tan^{-1}((13.8)/(4.6))=71.565\dots^\circ$$

So the interior angle ABC is

$$180^\circ-71.565\dots^\circ=108.434\dots^\circ$$

Answer: 108.4° .

- 8 The diagram shows an isosceles triangle.

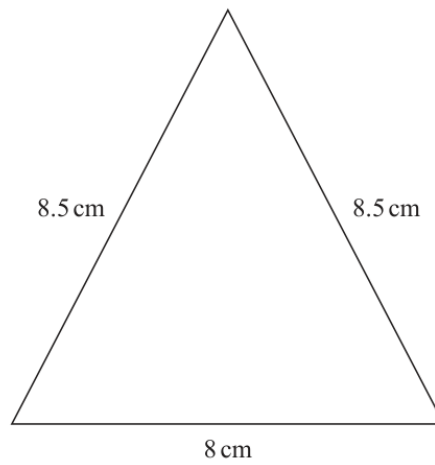


Diagram **NOT**
accurately drawn

Work out the area of the triangle.

.....cm²

(Total for Question 8 is 4 marks)

Answer reference | May-Nov 2020 | Question 8

Family: Use rectangle and triangle area • Variant: Recover a missing dimension from area

8	$8.5^2 - (8 \div 2)^2 (= 56.25)$ or $\cos x = \frac{4}{8.5}$ oe			M1 or eg $\cos A = \frac{8^2 + 8.5^2 - 8.5^2}{2 \times 8 \times 8.5}$
	$\sqrt{56.25}$ (= 7.5) or $x = \cos^{-1}\left(\frac{4}{8.5}\right)$ (= 61.927...) oe			M1 or eg $(A =) \cos^{-1}\left(\frac{8^2 + 8.5^2 - 8.5^2}{2 \times 8 \times 8.5}\right)$ (61.927...) (other angle = 56.144...)
	$8 \times "7.5" \div 2$ oe or $0.5 \times 8 \times 8.5 \times \sin "61.927..."$			M1 or eg $0.5 \times 8.5 \times 8 \times \sin "61.927..."$ oe
		30	4	A1
				Total 4 marks

Worked solution

May-Nov 2020 | Question 8 | 4 marks

Family: Use rectangle and triangle area • Variant: Recover a missing dimension from area

Official final mark-scheme pages are embedded immediately after this question.

- 4 The diagram shows rectangle $ABCD$

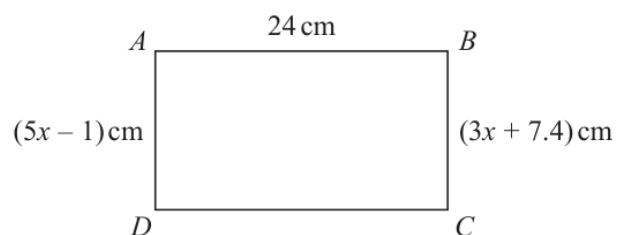


Diagram **NOT**
accurately drawn

Work out the perimeter of the rectangle.
Show your working clearly.

..... cm

(Total for Question 4 is 4 marks)

Answer reference | January 2023 | Question 4

Family: Find perimeter of polygonal shapes • Variant: Find perimeter of a simple triangle or rectangle

Question	working	Answer	Mark	Notes
4	eg $5x - 1 = 3x + 7.4$ oe or eg $10x - 2 + 48$ or $6x + 14.8 + 48$ or $24 + 24 + 5x - 1 + 3x + 7.4$ oe		4	M1 a correct equation to find x or a correct expression for the perimeter in terms of x
	$x = 4.2$			A1 the correct value of x (implies previous mark)
	$2 \times 24 + 2(5 \times "4.2" - 1)$ oe or $2 \times 24 + 2(3 \times "4.2" + 7.4)$ oe or $2 \times 24 + (5 \times 4.2 - 1) + (3 \times 4.2 + 7.4)$ oe eg $24 + 24 + 20 + 20$ oe			M1dep on a correct method to find the perimeter – use of positive x from correct working (1 st M1 awarded for an equation) and only if used the same measurement for AD and BC
	working required	88		A1 cao dep on either M1 or $x = 4.2$
				Total 4 marks

Worked solution

January 2023 | Question 4 | 4 marks

Family: Find perimeter of polygonal shapes • Variant: Find perimeter of a simple triangle or rectangle

Official final mark-scheme pages are embedded immediately after this question.